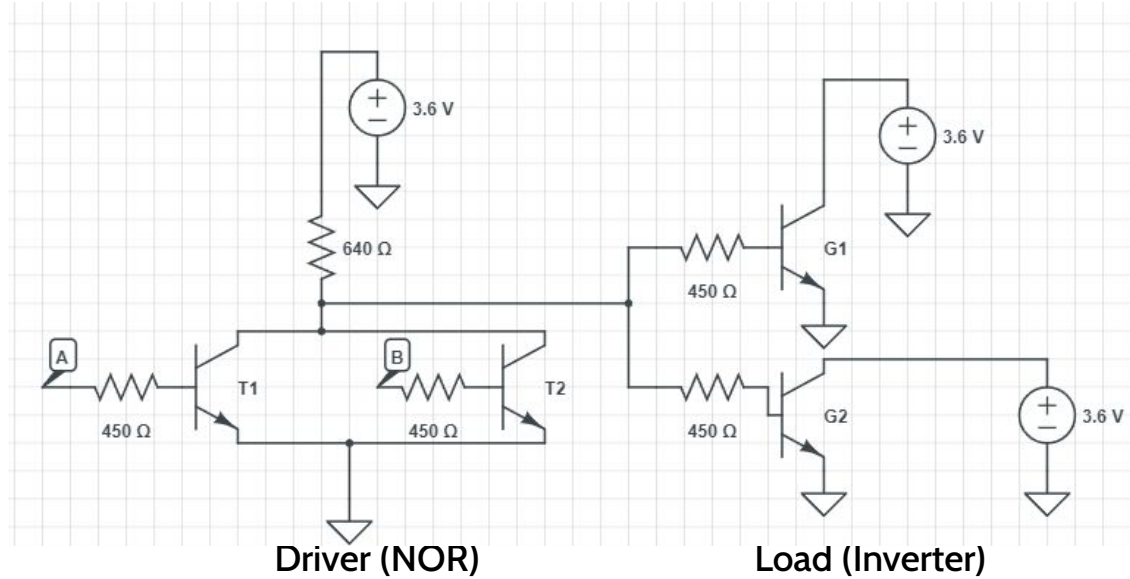


CSE350: Digital Electronics & Pulse Techniques

Lecture 05: RTL NOR, Intro to DTL

Solution to practice problem of Lec-04:



Let us first assume the driver circuit is *disconnected from the loads*. Assuming when input is high, the BJTs are in saturation & for low input, cutoff. We can construct the following table for the driver circuit:

When any one of the RTLs have high input, their BJTs are saturated & this sets the output voltage to 0.2V, which is the case for saturation mode. Only when both RTLs have low input, both BJTs are cutoff & thus there is no path for current from the source to flow. This open circuit leads to an output voltage of 3.6V.

A	B	T1	T2	V0
3.6	3.6	S	S	0.2
3.6	0	S	C	0.2
0	3.6	C	S	0.2
0	0	C	C	3.6

When we consider the driver is connected to the loads, **no change will occur for cases with low output**, since it is fixed for saturation mode that $V_{CE} = 0.2V$. However, **for the high output voltage case, it will reduce from 3.6V, since there will be path for current to flow due to connection with load**. So, we only need to analyze this case.

Now, the fanout is 5. So, we will consider 5 inverters connected to the output of the driver circuit. Since the output of the driver is high, all the inverters' input will also be high and lead them to **saturation**. This implies for all the loads, $V_{BE} = 0.8V$. The necessary part of the circuit for analysis will look like the following:

$$\text{Current through each input resistance of the loads} = \frac{V_o - 0.8}{460}$$

This will be equal to the current flowing through the 640 ohm resistance in

$$\text{driver circuit. So, } \frac{3.6 - V_o}{640} = \frac{V_o - 0.8}{450} * 5$$

$$\text{Solving, } V_o = 1.1452V$$

$$\text{Thus, base current of each load} = I_B = \frac{1.1452 - 0.8}{450} = 0.767mA$$

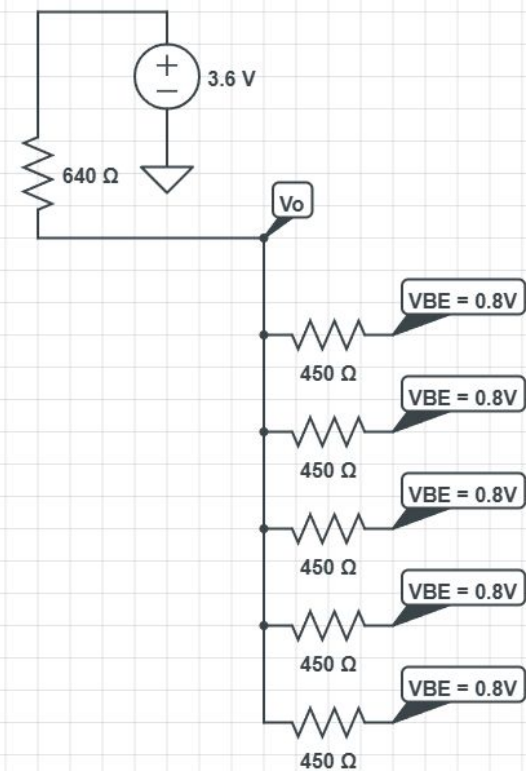
$$\text{Now, since the loads are in saturation, for all loads, } V_{CE} = 0.2V$$

$$\text{So, collector current of each load, } I_C = \frac{3.6 - 0.2}{640} = 5.3125mA$$

$$\text{This gives, } \beta_{forced} = \frac{I_C}{I_B}(\text{load}) = \frac{5.3125}{0.767} = 6.9263$$

This is the minimum value of β_F for the load inverters to remain in saturation.

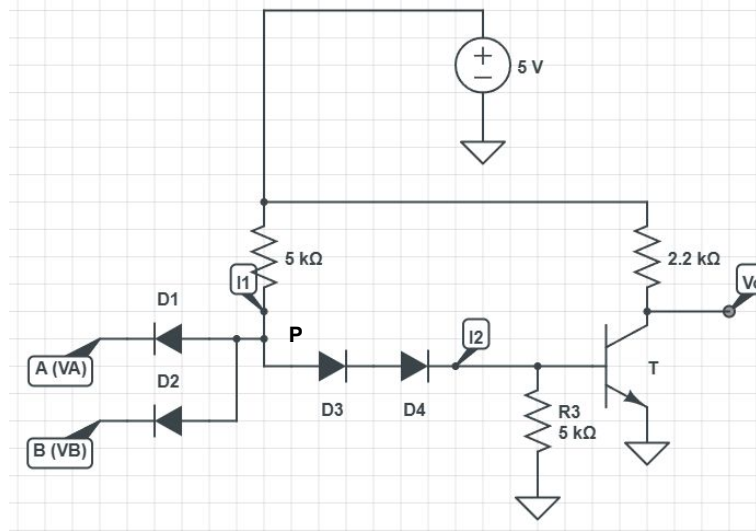
$$\beta_{F(min)} = \beta_{forced} = 6.9263$$



DTL (Diode Transistor Logic)

It is formed with **bipolar switching transistors** with diodes at their inputs (base). A switching transistor operates in cutoff & saturation mode.

DTL NAND Gate:



Note: The notations used in this ckt diagram are used in case (0,0) & case(0,1) only

Case(0,0): When both inputs are low, we may assume the anode voltages of the input diodes may be higher than the low cathode voltages. Thus the diodes will be conducting and we can replace them with 0.7V drops. However, *there is always a chance that the inputs to this NAND gate may actually be the outputs of some previous DTL circuit.* In such cases, **it is safe to assume that the low voltage will be 0.2 like in RTL ckt** instead of 0V.

Therefore, the anode voltage of the input diodes = $0.2 + 0.7 = 0.9V$

Now, we will show that the transistor T will remain in cutoff for this logic case. We know that the minimum voltage V_{BE} required to turn on the transistor is $0.5V$. Also, diodes D_3 & D_4 will need at least $0.6V$ across them when they are on. So, total voltage required at the anode of input diodes for the transistor to turn on = $0.6 \times 2 + 0.5 = 1.7V > 0.9V$. Thus, our assumption is correct.

Since, T is off, $I_B = I_C = I_E = 0$. Consequently, $V_o = 5V$

$$I_2 = 0 \quad I_1 = \frac{5 - 0.9}{5k} = 0.82mA \quad I_{D_1} = I_{D_2} = \frac{I_1}{2} = 0.41mA$$

Case(0,1)|(1,0): Both cases will give same result, so we will only analyze (0,1). Let's again assume the transistor T will be off.

Now, $V_B = 5V$ $V_A = 0.2V$. So, the diode at A will only be conducting. This can then be replaced by a $0.7V$ drop & through similar analysis as in case (0,0), we find that the anode voltage $V_P = 0.9V$.

Since the condition for turning on T is still the same, T will thus be off this time too.

So, again, $I_B = I_C = I_E = 0$. Consequently, $V_o = 5V$

$$I_2 = 0 \quad I_1 = \frac{5 - 0.9}{5k} = 0.82mA$$

However, this time,

$$I_{D_1} = I_1 = 0.82mA$$

$$I_{D_2} = 0$$

Case(1,1): This time, both diodes at the input will be off. Thus, the current from 5V, must flow through D_3 & D_4 . Thus both the diodes & transistor T must be on. Since, in DTL, a transistor is in saturation while on, we know, in saturation mode,

$$V_{BE} = 0.8V \text{ \& } V_{CE} = 0.2V$$

$$V_{D_4} = 0.7V \quad V_{D_3} = 0.7V$$

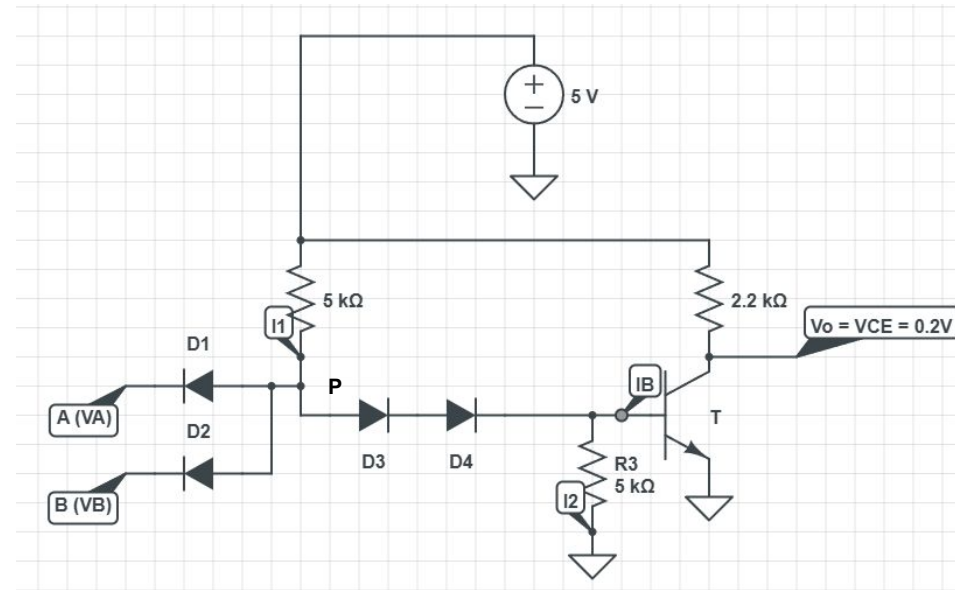
$$\text{So, } V_P = V_{D_3} + V_{D_4} + V_{BE} = 0.7 + 0.7 + 0.8 = 2.2V$$

Thus, $I_{D_1} = I_{D_2} = 0$ $I_1 = \frac{5 - 2.2}{5k} = 0.56mA$ $I_2 = \frac{0.8}{5k} = 0.16mA$

$$I_B = I_1 - I_2 = 0.4mA$$

$$I_C = \frac{5 - V_o}{2.2k} = \frac{5 - V_{CE}}{2.2k} = 2.182mA$$

Note: The notations used in this ckt diagram are used use in case(1,1).



Let us form a table of the cases and corresponding results:

A	B	V_A	V_B	T	V_o
0	0	0.2	0.2	C	5
0	1	0.2	5	C	5
1	0	5	0.2	C	5
1	1	5	5	S	0.2

Additionally, let us calculate the minimum value of β that will keep the transistor T operating in the correct modes for all logic cases. We see that T is in cutoff in all cases except in case (1,1), in which it is in saturation. So, to keep T in saturation for high voltage inputs to both terminals, we must ensure the forward β_F is greater than the existing β_{forced} of saturation mode.

We get, $\beta_{forced} = \frac{I_C}{I_B} = \frac{2.182}{0.4} = 5.46$

So, $\beta_{F(min)} = \beta_{forced} = 5.46$